

Chapter 1 Assignment

Student: UWIZEYE Docile

Student ID: 27549

Course: Introduction to Big Data with Python

Assignment: Chapter 1 - Environment Setup and Python Basics

Task 1: Student Profile System (20 marks)

```
In [1]: # Task 1: Student Profile System
# Collect student information and display a formatted profile

print('==== STUDENT PROFILE SYSTEM ====')

# Student data (hardcoded for demonstration)
full_name      = 'UWIZEYE Docile'
student_id     = '27549'
age            = 22
country        = 'Rwanda'
year_of_study  = 3
gpa            = 3.75
major          = 'Software Engineering'

# Calculate expected graduation year
current_year    = 2025
remaining_years = 4 - year_of_study
graduation_year = current_year + remaining_years

# Display formatted student profile using f-strings
print()
print('=' * 45)
print('          STUDENT PROFILE CARD')
print('=' * 45)
print(f' Full Name      : {full_name}')
print(f' Student ID     : {student_id}')
print(f' Age            : {age} years')
print(f' Country        : {country}')
print(f' Year of Study   : Year {year_of_study}')
print(f' GPA            : {gpa:.2f}')
print(f' Major          : {major}')
print(f' Graduation Year : {graduation_year}')
print('=' * 45)
```

===== STUDENT PROFILE SYSTEM =====

```
=====
                        STUDENT PROFILE CARD
=====
Full Name       : UWIZEYE Docile
Student ID      : 27549
Age             : 22 years
Country         : Rwanda
Year of Study   : Year 3
GPA             : 3.75
Major           : Software Engineering
Graduation Year : 2026
=====
```

Task 2: Advanced Calculator (5 marks)

```
In [2]: # Task 2: Advanced Calculator
# Perform all arithmetic operations on two numbers

print('===== ADVANCED CALCULATOR =====')

# Two numbers for demonstration
num1 = 64.0
num2 = 9.0

print()
print('=' * 40)
print('      ARITHMETIC OPERATIONS')
print('=' * 40)
print(f' Addition       : {num1} + {num2} = {num1 + num2:.2f}')
print(f' Subtraction    : {num1} - {num2} = {num1 - num2:.2f}')
print(f' Multiplication  : {num1} x {num2} = {num1 * num2:.2f}')
print(f' Division        : {num1} / {num2} = {num1 / num2:.4f}')
print(f' Floor Division  : {num1} // {num2} = {num1 // num2:.2f}')
print(f' Modulus         : {num1} % {num2} = {num1 % num2:.2f}')
print(f' Exponentiation  : {num1} ^ {num2} = {num1 ** num2:.2f}')

print()
print('      EVEN OR ODD CHECK')
print('=' * 40)
num1_type = 'Even' if int(num1) % 2 == 0 else 'Odd'
num2_type = 'Even' if int(num2) % 2 == 0 else 'Odd'
print(f' {num1} is {num1_type}')
print(f' {num2} is {num2_type}')

print()
print('      PERCENTAGE')
print('=' * 40)
percentage = (num1 / num2) * 100
print(f' {num1} is {percentage:.2f}% of {num2}')
print('=' * 40)
```

===== ADVANCED CALCULATOR =====

=====

ARITHMETIC OPERATIONS

=====

```
Addition      : 64.0 + 9.0 = 73.00
Subtraction    : 64.0 - 9.0 = 55.00
Multiplication : 64.0 x 9.0 = 576.00
Division       : 64.0 / 9.0 = 7.1111
Floor Division : 64.0 // 9.0 = 7.00
Modulus        : 64.0 % 9.0 = 1.00
Exponentiation : 64.0 ^ 9.0 = 18014398509481984.00
```

EVEN OR ODD CHECK

=====

```
64.0 is Even
9.0 is Odd
```

PERCENTAGE

=====

```
64.0 is 711.11% of 9.0
```

=====

Task 3: Data Type Explorer (5 marks)

In [3]: *# Task 3: Data Type Explorer*

```
print('==== DATA TYPE EXPLORER ====')
print()

my_int    = 25
my_float  = 3.14
my_str    = 'Big Data is transforming the world today'
my_bool   = True
my_none   = None

print('--- Original Types ---')
print(f' my_int    = {my_int}    -> type: {type(my_int)}')
print(f' my_float  = {my_float}  -> type: {type(my_float)}')
print(f' my_str    = "{my_str}"    -> type: {type(my_str)}')
print(f' my_bool   = {my_bool}    -> type: {type(my_bool)}')
print(f' my_none   = {my_none}    -> type: {type(my_none)}')

print()
print('--- Type Conversions ---')
print(f' int -> float : {float(my_int)} type: {type(float(my_int))}')
print(f' int -> str   : "{str(my_int)}" type: {type(str(my_int))}')
print(f' int -> bool  : {bool(my_int)} type: {type(bool(my_int))}')
print(f' float -> int : {int(my_float)} type: {type(int(my_float))}')
print(f' float -> str : "{str(my_float)}" type: {type(str(my_float))}')
print(f' bool -> int  : {int(my_bool)} type: {type(int(my_bool))}')
print(f' bool -> str  : "{str(my_bool)}" type: {type(str(my_bool))}')
num_str = '42'
print(f' str -> int   : {int(num_str)} type: {type(int(num_str))}')
print(f' str -> float : {float(num_str)} type: {type(float(num_str))}')

print()
print('--- String Methods on my_str ---')
print(f' Original    : {my_str}')
```

```

print(f' upper()      : {my_str.upper()}')
print(f' lower()      : {my_str.lower()}')
print(f' title()       : {my_str.title()}')
print(f' replace()    : {my_str.replace("Big Data", "Python")}')
print(f' split()       : {my_str.split()}')
print(f' count("a")    : {my_str.count("a")}')
print(f' find("world"): {my_str.find("world")}')
print(f' startswith   : {my_str.startswith("Big")}')
print(f' strip()       : {" hello ".strip()}')

print()
print('--- String Indexing on my_str ---')
print(f' First character (index 0) : {my_str[0]}')
print(f' Last character (index -1) : {my_str[-1]}')
print(f' Character at index 3       : {my_str[3]}')
print(f' Characters index 2 to 7    : {my_str[2:7]}')

```

===== DATA TYPE EXPLORER =====

--- Original Types ---

```

my_int   = 25    -> type: <class 'int'>
my_float = 3.14  -> type: <class 'float'>
my_str   = "Big Data is transforming the world today" -> type: <class 'str'>
my_bool  = True  -> type: <class 'bool'>
my_none  = None  -> type: <class 'NoneType'>

```

--- Type Conversions ---

```

int -> float : 25.0  type: <class 'float'>
int -> str   : "25"  type: <class 'str'>
int -> bool  : True  type: <class 'bool'>
float -> int : 3     type: <class 'int'>
float -> str : "3.14" type: <class 'str'>
bool -> int  : 1     type: <class 'int'>
bool -> str  : "True" type: <class 'str'>
str -> int   : 42    type: <class 'int'>
str -> float : 42.0  type: <class 'float'>

```

--- String Methods on my_str ---

```

Original      : Big Data is transforming the world today
upper()       : BIG DATA IS TRANSFORMING THE WORLD TODAY
lower()       : big data is transforming the world today
title()       : Big Data Is Transforming The World Today
replace()     : Python is transforming the world today
split()       : ['Big', 'Data', 'is', 'transforming', 'the', 'world', 'today']
count("a")    : 4
find("world") : 29
startswith    : True
strip()       : "hello"

```

--- String Indexing on my_str ---

```

First character (index 0) : B
Last character (index -1) : y
Character at index 3      :
Characters index 2 to 7   : g Dat

```

Task 4: String Processing Pipeline (5 marks)

In [4]: *# Task 4: String Processing Pipeline*

```
raw_data = '    MUGISHA Jean Pierre | 1998 | Kigali, Rwanda | Computer Science |'

cleaned = raw_data.strip()
print(f'Stripped: {cleaned}')

fields = cleaned.split('|')
fields = [field.strip() for field in fields]
print(f'Fields: {fields}')

full_name_raw = fields[0]
birth_year = int(fields[1])
location = fields[2]
major = fields[3]
gpa = float(fields[4])

full_name = full_name_raw.title()
name_parts = full_name.split()
last_name = name_parts[0]
first_name = ' '.join(name_parts[1:])
full_name_combined = first_name + ' ' + last_name

current_year = 2025
age = current_year - birth_year

location_parts = location.split(',')
city = location_parts[0].strip()
country = location_parts[1].strip()

has_science = 'Science' in major

print()
print('=' * 40)
print('    EXTRACTED PROFILE SUMMARY')
print('=' * 40)
print(f' Full Name      : {full_name}')
print(f' First Name     : {first_name}')
print(f' Last Name      : {last_name}')
print(f' Full Name      : {full_name_combined}')
print(f' Birth Year      : {birth_year}')
print(f' Age            : {age} years')
print(f' City           : {city}')
print(f' Country        : {country}')
print(f' Major          : {major}')
print(f' GPA            : {gpa}')
print(f' Has Science    : {has_science}')
print('=' * 40)
```

Stripped: MUGISHA Jean Pierre | 1998 | Kigali, Rwanda | Computer Science | 3.72
 Fields: ['MUGISHA Jean Pierre', '1998', 'Kigali, Rwanda', 'Computer Science', '3.72']

```
=====
EXTRACTED PROFILE SUMMARY
=====
Full Name      : Mugisha Jean Pierre
First Name     : Jean Pierre
Last Name      : Mugisha
Full Name      : Jean Pierre Mugisha
Birth Year      : 1998
Age            : 27 years
City           : Kigali
Country        : Rwanda
Major          : Computer Science
GPA            : 3.72
Has Science    : True
=====
```

Task 5: Financial Calculator (5 marks)

```
In [5]: # Task 5: Financial Calculator

print('==== FINANCIAL CALCULATOR ====')

# Sample values for demonstration
principal      = 500000
annual_rate     = 12.0
years          = 3
monthly_expense = 20000.0

# Simple Interest:  $I = P \times R \times T / 100$ 
simple_interest = principal * annual_rate * years / 100
simple_total    = principal + simple_interest

# Compound Interest:  $A = P \times (1 + R/100)^T$ 
compound_total = principal * (1 + annual_rate / 100) ** years
compound_interest = compound_total - principal

interest_difference = compound_total - simple_total
total_expenses      = monthly_expense * 12 * years
net_savings          = compound_total - total_expenses

print()
print('=' * 50)
print('          FINANCIAL SUMMARY REPORT')
print('=' * 50)
print(f'Principal Amount      : {principal:>15,.2f} RWF')
print(f'Annual Rate           : {annual_rate:>14.2f}%')
print(f'Period                : {years:>12} years')
print(f'Monthly Expense       : {monthly_expense:>15,.2f} RWF')
print('-' * 50)
print(f'Simple Interest        : {simple_interest:>15,.2f} RWF')
print(f'Total (Simple)         : {simple_total:>15,.2f} RWF')
print('-' * 50)
print(f'Compound Interest     : {compound_interest:>15,.2f} RWF')
print(f'Total (Compound)      : {compound_total:>15,.2f} RWF')
print('-' * 50)
```

```

print(f' Interest Difference      : {interest_difference:>15,.2f} RWF')
print(f' Total Expenses          : {total_expenses:>15,.2f} RWF')
print(f' Net Savings             : {net_savings:>15,.2f} RWF')
print('=' * 50)

```

===== FINANCIAL CALCULATOR =====

```

=====
                        FINANCIAL SUMMARY REPORT
=====
Principal Amount      :      500,000.00 RWF
Annual Rate           :           12.00%
Period                :           3 years
Monthly Expense       :      20,000.00 RWF
-----
Simple Interest       :      180,000.00 RWF
Total (Simple)        :      680,000.00 RWF
-----
Compound Interest    :      202,464.00 RWF
Total (Compound)     :      702,464.00 RWF
-----
Interest Difference   :       22,464.00 RWF
Total Expenses        :      720,000.00 RWF
Net Savings           :      -17,536.00 RWF
=====

```